

JOSEPHINE BURKE

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SUMMARY

Final-year Computer Science, Data Science & AI student who bridges technical systems and business outcomes – prototyping data solutions directly with client and executive stakeholders across finance, federal, and applied-AI settings. Turns messy, high-stakes data into decisions.

EDUCATION

University College Dublin

Dublin, Ireland

B.Sc. Computer Science with Data Science and Artificial Intelligence

Expected May 2027

- **Relevant Coursework:** Machine Learning, Generative AI: Language Models, Data Mining, Cloud Computing, Quantum Machine Learning; Data Science in Python, Programming for Big Data, Information Visualisation, Network Analysis, Intro to AI, Probability
- Technical Officer, AI in Medicine Society - manage technical requirements and events

EXPERIENCE

Scotiabank - Global Banking & Markets

Dublin, Ireland

Technology Summer Intern

June 2026 - Aug 2026

- Partnered with Front Office trading desks across Dublin and London to scope a system-dependency mapping initiative, translating trader workflows into technical requirements.
- Rapidly prototyped a full-stack trading analytics dashboard (**React/TypeScript**) on **ION MMI** data, empowering traders to self-serve venue, price, yield, and client-level views.
- Built a SQL deduplication pipeline (**Dremio**) collapsing duplicate trade IDs onto one economic trade, improving reporting accuracy and cutting manual reconciliation; automated deck generation via **Excel VBA** for the Dublin team.

Accure, Inc.

Remote

Computer Science & Data Engineering Intern

Mar 2026 - June 2026

- Served as main point of contact, coordinating requirements directly with the client's **CEO, General Counsel**, and technical team.
- Led a team of interns to rapidly prototype a full-stack marketplace (Stripe checkout, provider onboarding, revenue splits) for a humanitarian nonprofit.

United States Marshals Service

Arlington, Virginia

Data Engineering Intern - Investigative Operations Division, Behavioral Analysis Unit

June 2025 - Aug 2025

- Rapidly prototyped a Python (Dash/Plotly) investigative dashboard turning thousands of case files into an interactive tool, empowering investigators to self-serve analysis and cutting review time.
- Built NLP pipelines (spaCy, regex) to extract entities and drive network-graph and geospatial visualizations.
- Obtained a **U.S. Public Trust** designation through the internship's federal background investigation.

Quantum Computing Research

Remote

Research Intern

July 2022 - Aug 2022

- Designed and ran quantum circuits on IBM's Quantum Platform with Prof. Sebastian Deffner (UMBC); presented research on entanglement-assisted invariance.

UNIVERSITY PROJECTS

Mapping the Missing - Data Science in Practice | *Python, Pandas, GeoPandas, scikit-learn*

- Fused a ~26K-case NamUs scrape with 14 demographic, crime, and geospatial datasets to model county-level missing-persons patterns.
- Validated six risk signals via LASSO and quantified a $4.15\times$ over-representation of Indigenous cases.

Cyclistic Distributed Data Analysis - Programming for Big Data | *Python, Apache Spark, SQL*

- Processed ~14.8M bike-share trips (2.89 GB) via an automated ingestion pipeline (S3 download, schema normalization, weather join).
- Re-implemented bottleneck aggregations in Apache Spark (MapReduce), achieving 72–78% query speedups.

Dublin Bikes - Network Analysis | *Python, NetworkX, Gephi*

- Modeled the bike-share system as a directed weighted graph (109 stations, 1,187 edges); used centrality metrics to identify hub and bottleneck stations.
- Ran temporal network analysis to flag stations whose load imbalance exceeds physical dock capacity.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, SQL, HTML/CSS

Concepts: Data pipelines, data deduplication, knowledge graphs, NLP / entity resolution, big data & MapReduce

Data & ML: Pandas, NumPy, scikit-learn, GeoPandas, Apache Spark, NetworkX, spaCy

Tools & Platforms: Git, Dremio, React, Dash, Plotly, Stripe API, Excel VBA; MySQL, PostgreSQL

CERTIFICATIONS & ACTIVITIES

Accure AI: End-to-End No-Code AI (Data Engineering & ML); Mastering Generative AI (AI Agents, LLMs) | **The Coding School:** Intro to AI; Intro to Quantum Computing | **Secure Code Warrior:** Secure Coding Certification | **HackMIT 2026:** Selected Participant, MIT's flagship collegiate hackathon